

Molcontroller: A VMD Graphical User Interface Featuring Molecule Manipulation

ChenChen Wu, Shengtang Liu, Shitong Zhang, and Zaixing Yang*



Cite This: *J. Chem. Inf. Model.* 2020, 60, 5126–5131



Read Online

ACCESS |



Metrics & More



Article Recommendations

ABSTRACT: Visual Molecular Dynamics (VMD) is one of the most widely used molecular graphics software in the community of theoretical simulations. So far, however, it still lacks a graphical user interface (GUI) for molecular manipulations when doing some modeling tasks. For instance, translation or rotation of a selected molecule(s) or part(s) of a molecule currently only can be achieved using *tcl* scripts. Here, we use the *Tcl/Tk* toolkit to develop a user-friendly GUI for VMD, named *Molcontroller*, which is featured by allowing users to quickly and conveniently perform various molecular manipulations. This GUI might be helpful for improving the modeling efficiency of VMD users.



I. INTRODUCTION

Visual Molecular Dynamics (VMD)¹ is one of the most powerful molecular visualization and modeling programs for the researchers of theoretical simulations. However, there is still a lack of a graphical user interface (GUI) for molecular manipulations in the instances of doing some modeling tasks, such as translation/rotation of a selected molecule(s) or part(s) of a molecule or merging two or even more molecules into a single file, which, currently, can only be achieved using *tcl* scripts. In light of this, we develop a user-friendly GUI for VMD (based on version 1.9.3), named *Molcontroller*, to perform these daily-used manipulations. Moreover, it can also provide a variety of molecular identifier information in the VMD display window, which should be very convenient for briefly displaying the molecular structure features. This GUI might be helpful for improving the modeling efficiency of VMD users.

II. TECHNICAL IMPLEMENTATION AND MOLCONTROLLER FEATURES

The *Molcontroller*-GUI is developed using the *Tcl/Tk* toolkit and can be integrated into the *Extensions/Modeling/Molcontroller* menu of VMD (find more details in the “*Installation*” guide section). This GUI contains several function boxes, for instance, *Information*, *Property*, *Manipulation*, and *Save File* (Figure 1). The major features are described as follows.

II.a. Information. There are three panels in the *Information* box. In the left panel (*File List*), the targeting molecule(s) for modeling manipulations is listed. In the middle panel (*Unit Cell*), the user can switch the simulation box of the selected

molecule on or off using the radio button. Moreover, the user can also directly modify the VMD unit-cell parameters, i.e., *a*, *b*, *c*, *alpha*, *beta*, and *gamma*. The logging control panel is located in the right region of *Information* box. When the user clicks on the “*logging on*” radio button, the save file dialogue box is provided to allow the user to define the logging file. Then, the rotating and moving manipulations will be automatically recorded into the preset logging file. The manipulation recording process will stop if the user clicks on the “*logging off*” radio button. The users can noninteractively replay manipulations by loading the previously saved log file under *File/Load Visualization* in the VMD menu.

II.b. Property. The *Property* box can display various molecular identifier information. The first three columns of the *Property* box, from *Chain* >> *Residue* >> *Atom*, display the molecular information in a tree-based structure. The column of *Res_Count* displays the total number of each type of residue in a selected molecule. If a specific residue is clicked in the column of *Res_Count*, all residual IDs of the same residue will be listed in the column of *Res_Location*. The user can click the selections in the *Property* box to display the entire or some specific parts of a molecule. Alternatively, this can also be achieved by typing the text commands of VMD in the *Selected*

Received: July 3, 2020

Published: September 15, 2020



ACS Publications

© 2020 American Chemical Society

5126

<https://dx.doi.org/10.1021/acs.jcim.0c00754>
J. Chem. Inf. Model. 2020, 60, 5126–5131

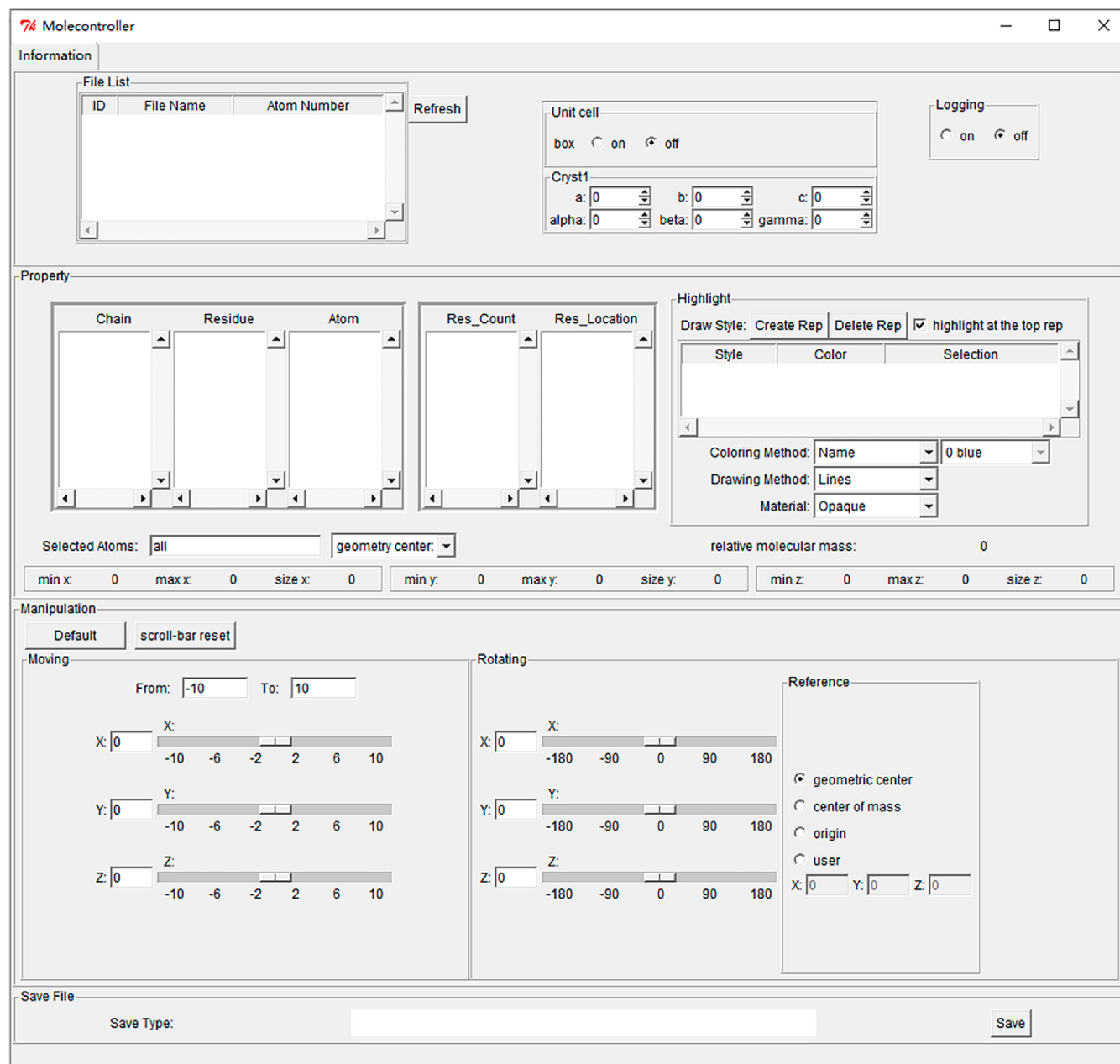


Figure 1. Main window of the *Molcontroller* graphical user interface. It contains four function boxes: *Information*, *Property*, *Manipulation*, and *Save File*.

Atoms text entry. In the box of *Highlight*, the user can render the selected molecule (or its part(s)) using the routine drawing style and coloring method of VMD software on the topmost representation by clicking the *highlight at the top rep* checkbox. The geometric center or center of mass (COM), relative molecular mass, and minimum enclosing rectangle dimension (range and size) of the selected molecule (or its part) are displayed at the bottom of the *Property* box. In addition, the number of each type of residue in the protein can be counted and saved directly (more below).

II.c. Manipulation. The main function of the *Manipulation* box is to implement the translation or rotation of a selected molecule (or its part(s)). The moving range can be defined in the text entry boxes labeled with *From* and *To*. The selected molecule(s) or selection(s) of a molecule can be translated and rotated along the specified *x*-, *y*-, or *z*-axis with preset values

that are declared in the *Moving* and *Rotating* text entry boxes, respectively. Alternatively, these two operations can also be done through dragging the specific slider(s) along the *x*-, *y*-, or *z*-axis. The rotation reference point of the molecule can be set to the geometric center or the COM of the molecule, the origin of the system, or a user-defined reference point using the radio button. The *Scroll bar reset* button can reset the moving slider to the initial value for further adjustments. The *Default* button can restore the original structure of the molecule for the user to readjust its structure.

II.d. Save File. The last box of the GUI is designed to save the current molecule or selection(s) of a molecule after moving or rotating operation(s) or some structural analysis results (e.g., the number of each type of residue). The user can also merge these processed molecules into a single PDB file and

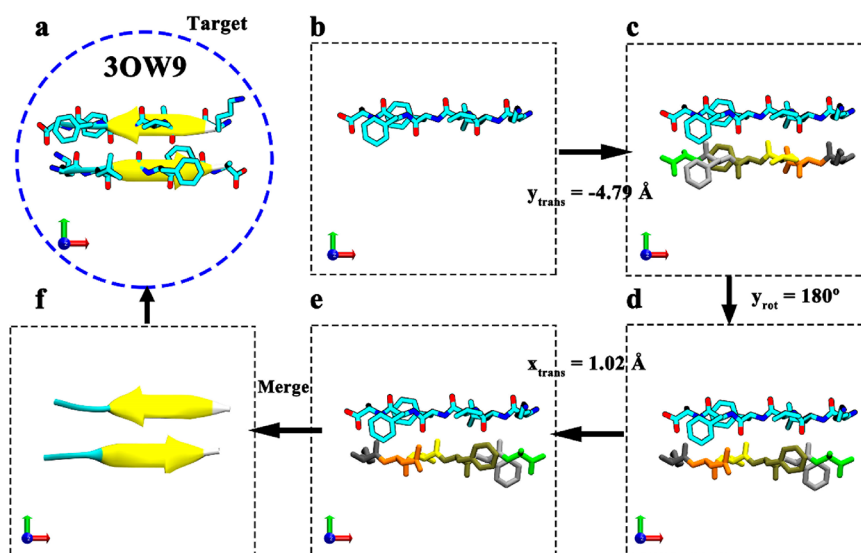


Figure 2. (a) Targeting configuration of the KLVFFA dimeric antiparallel β -sheet (PDB ID: 3OW9) shown as the Newcartoon model and colored with the secondary structure. (b–f) Schematic of manipulation process of building KLVFFA dimeric antiparallel β -sheet starting with two identical KLVFFA peptides. All the translation and rotation parameters are obtained from the crystal structure (PDB ID: 3OW9).

save them as a complex. The file types for saving are available in the drop-down menu at the bottom of the *Save File* box.

III. SPECIFIC EXAMPLES

III.a. Peptide Manipulations. To show the molecular manipulations feature of our GUI, we simply construct an antiparallel β -sheet consisting of two $A\beta_{16-21}$ peptides (KLVFFA), which is the core amyloidogenic segment of the amyloid fibril in Alzheimer's disease. The ideal targeting configuration for constructing is the KLVFFA β -sheet crystal structure that has been deposited in the protein data bank with the PDB ID of 3OW9² (Figure 2a). All the translation and rotation parameters during the manipulation process are derived from the KLVFFA β -sheet crystal structure. First, two identical KLVFFA peptides are loaded into VMD. Then, upon opening the *Molcontroller*-GUI and clicking the *Refresh* button in the *Information* panel, the two peptides are loaded in the GUI. Meanwhile, their file IDs, file names, and atom numbers are also listed in the *File List* box. Obviously, the initial coordinate of the two peptides is completely overlapped (Figure 2b), which means that to build an antiparallel β -sheet structure, one of the peptides needs to be translated and rotated accordingly. Hence, a peptide is selected and translated 4.79 Å along with the negative direction of the y -axis (Figure 2c). Then, the COM of this peptide is assigned as the rotation reference point, and the peptide is rotated 180° along the y -axis (Figure 2d). Furthermore, it is translated 1.02 Å along the x -axis (Figure 2e). Finally, the two peptides are merged into a single PDB file completing the manipulation task (Figure 2f).

III.b. Protein Feature Display and Structural Analysis.

To show the features of interactive visualization and fundamental molecular structure analysis of this GUI, we use a typical potassium-channel protein, KcsA (PDB ID: 2K1E),³ as an example (Figure 3). First, the PDB file of KcsA is loaded through the *Molecule File Browser*. Then, when opening the *Molcontroller*-GUI and clicking the *Refresh* button in the *Information* panel, users can find that the file ID, file name, and atom number of the protein in the PDB file are listed in the *File List* box. Meanwhile, in the property box, much molecular

identifier information is available. For example, in the *chain* column, it can be clearly seen that KcsA contains four chains, from chain A to chain D. When clicking chain A, one can find that all residues in chain A are listed in the *Residue* column. When users click any residue in chain A in the *Residue* column, all atoms of the selected residue and the total number of each type of residue will be listed in the *Atom* column and *Res_count* column, respectively (Figure 3a). When an exact type of residue is selected, all residual IDs of this type of residue will be listed in the *Res_Location* column. Here, chain A, chain B, chain C, and chain D are shown with Newcartoon style and colored with orange, yellow, cyan, and tan, respectively. All Arg residues are selected and highlighted with van der Waals (vdW) ball style for the subsequent structural analysis (Figure 3b). In addition, the total number of each type of residue in the protein can also be analyzed (Figure 3c). Clearly, plentiful molecular identifier information integrated into this GUI endows researchers more convenience when they perform analyzing tasks on some protein structural characteristics.

III.c. Multichain Protein Manipulation. The manipulation of the position of ions or lipids for the multichain KcsA protein is fundamental but quite important for building the initial conformation in MD simulations, which is frequently involved in studies on the potassium ion channel.^{4,5} There are various Web servers or molecular dynamics simulation packages that offer an interface for inserting a protein into membrane lipids.^{6,7} Among them, the CHARMM graphical user interface⁸ (CHARMM-GUI) is representative and widely employed as a powerful and versatile web-based graphical user interface for building complex systems, including the generation of the membrane-embedded protein model. The position and orientation of the protein can be automatically determined in a built-in algorithm within the CHARMM-GUI according to the user's different choices. However, the CHARMM-GUI has some limitations on user-interaction capabilities and real-time responsiveness in the process of building protein/membrane complexes. Specifically, the users cannot view in real-time the position and orientation of the protein in the visualization window after performing a manipulation (orient the protein step of "CHARMM-GUI →

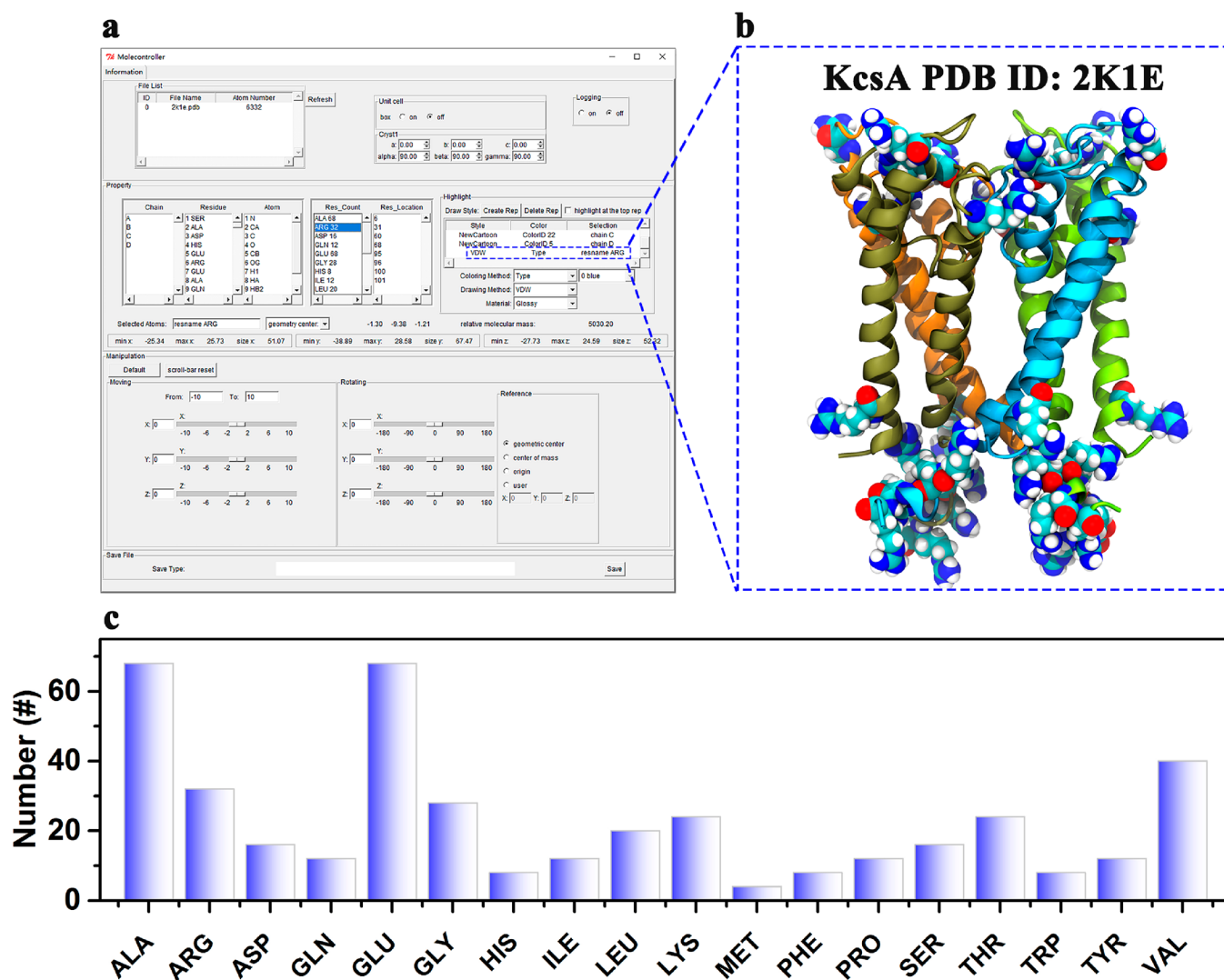


Figure 3. (a) Structural analysis of the KcsA protein using Molcontroller. (b) Chain A, chain B, chain C, and chain D are drawn with Newcartoon style and colored with orange, yellow, cyan, and tan, respectively. All 32 arginine residues are highlighted with vdW balls. (c) Total number of each type of residue in the KcsA protein.

Membrane Builder → Bilayer Builder”). Here, we built a membrane-KcsA model using the Molcontroller-GUI via five steps (Figure 4) and compared the model with that created by the CHARMM-GUI. It was found that the position, orientation, and other structural properties of the two models are in good accordance with each other. More details are shown below.

Step 1. Construction of protein/membrane complexes. The 1-palmitoyl-2-oleoylphosphatidylcholine (POPC) membrane with a size of $100 \text{ \AA} \times 100 \text{ \AA}$ in the x - y plane was generated using the VMD built-in *Membrane Builder Plugin* (CHARMM36 force field was chosen, and water shells were removed). The created file was saved as a single PDB file. The KcsA protein (PDB ID: 2K1E) and POPC membrane were loaded into VMD by clicking File → New Molecule. Then, the user can open the Molcontroller-GUI and merge them into a single PDB file for subsequent processing (Figure 4a).

Step 2. Manipulation of the protein conformation. The spatial position parameters of the KcsA protein with respect to the membrane can be defined as its shift (d), rotational (α), and tilt angle (β) along the membrane normal direction. These parameters can be obtained by experimental characterizations

or theoretical calculations. Here, the value of each parameter of KcsA is estimated according to the Orientation of Proteins in Membranes database (OPM, <https://opm.phar.umich.edu/proteins/59>),⁹ which is widely adopted in membrane-embedded protein studies. Specifically, the parameters in this model are $d(z_{\text{trans}}) \approx 3 \text{ \AA}$, $\alpha(x_{\text{rot}}) \approx 90^\circ$, $\alpha(y_{\text{rot}}) \approx -5^\circ$, and $\beta \approx 0^\circ$, where the rotation reference point is the COM of the protein (Figure 4b). Usually, these manipulations can be automatically completed in a built-in algorithm within the CHARMM-GUI by selecting different orientation options (“CHARMM-GUI → Membrane Builder → Bilayer Builder”, Orientation step). However, it is hard to access the structure in real-time after manipulation, and the user cannot promptly judge whether there are unwanted mistakes in the model or further manipulate protein position and orientation in this step with special requirements. For the Molcontroller-GUI, this manipulation can be done through inputting parameters by textboxes or dragging the sliding bar in the Manipulation box, which offers a much more flexible manner for generating membrane/protein complex models.

Step 3. Removing lipids that overlap with the protein. Because there are overlapping spatial positions between lipids

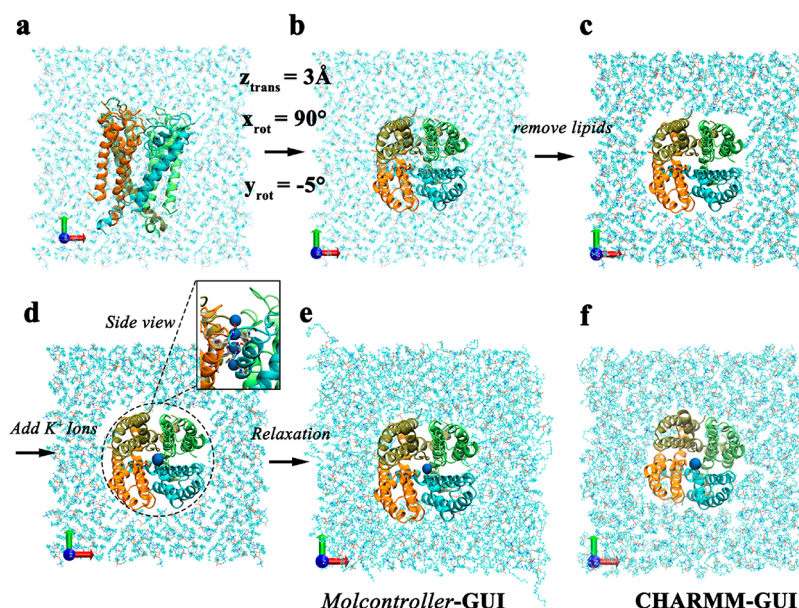


Figure 4. (a) Combined structure of the POPC membrane and KcsA protein; the same color scheme used in Figure 3b is assigned to the KcsA protein. (b) The translation and rotation of KcsA in the POPC membrane to adjust the position and orientation of the ion channel. (c) The structure after the removal of POPC lipids overlapping with the KcsA protein. (d) The insertion of potassium ions into the KcsA protein channel around the residue TVGYG. The side view is provided in the upper right corner. (e) and (f) are the conformation comparison between the results of the Molcontroller-GUI and CHARMM-GUI.

and protein, it is necessary to remove lipids that overlap with the protein, and we follow the protocol in Aksimentiev et al.¹⁰ More specifically, most of the molecules that collide with the KcsA protein were removed using the VMD selected atoms command, “not (same residue as rename POPC and $(x\wedge 2 + y\wedge 2) < 400$)”. It is noteworthy that the KcsA protein is cylindrical-like, and lipids within the cylinder should be selected and deleted. Here, a cylinder radius of 20 Å is large enough to cover most of the overlapping lipids, though it is smaller than the actual radius of the KcsA protein (24 Å). The final conformation is shown in Figure 4c.

Step 4. Addition of potassium ions. Given that there are four evenly spaced potassium ion (K^+) binding sites around residues 54–58 (TVGYG) in the channel of the KcsA protein, we first generated four identical K^+ ions at the geometry center of the binding sites and subsequently manipulated them into the suitable position by using the moving control button in the Molcontroller-GUI, as shown in Figure 4d. The geometry center of residue TVGYG, (0.62, −1.18, 11.53), can be obtained from the Property box when a selection command “protein and residue 54 to 58” is sent to the Selected Atoms text entry.

Step 5. Relaxation of lipids. After the model construction, we can relax the system using molecular dynamics (MD) simulations performed by several popular packages, such as NAMD and GROMACS.^{11,12} Here, the created model was minimized by the steepest descent method and then subjected to a 40 ps NPT ensemble pre-equilibration process using the GROMACS package (version 2019, the protein was restrained).¹² The CHARMM36 force field¹³ and TIP3P water model¹⁴ were used in this simulation. The 0.15 mol/L NaCl ions were added to neutralize the protein net charge. A comparison between conformations generated by the CHARMM-GUI and our results is shown in Figure 4d and 4e. The position, orientation, and other structural character-

istics of the KcsA protein and POPC membrane are in good agreement with the CHARMM-GUI results.

IV. INSTALLATION (REQUIRES AT LEAST VMD VERSION 1.9.3)

The plugin can be added to the Extensions submenu in the VMD menu bar. To install Molcontroller on Windows, Linux, or MacOS, follow the following steps:

1. Create a personal plugin directory, recommended as follows:

VMD-Installation-Directory/plugins/noarch/tcl/molcontroller1.0, where VMD-Installation-Directory is the VMD installation directory.

2. Download the scripts; archive and extract it in the above directory:

The source code is released under the GNU¹⁵ public license version 3 from <https://github.com/ChenchenWu-hub/Molcontroller>.

3. Add the following line to your configuration file of VMD (e.g., VMD-Installation-Directory/vmd/vmd.rc for Windows, VMD-Installation-Directory/lib/vmd/.vmdrc for Linux, and VMD-Installation-Directory/Contents/vmd/.vmdrc for MacOS, respectively):

```
lappend auto_path VMD-Installation-Directory/plugins/noarch/tcl/molcontroller1.0
vmd_install_extension molcontrol molcontroller_tk
“Modeling/Molcontrol”
```

If the plugin installation is successful, you will see the plugin under Extensions/Modeling/Molcontroller in the VMD menu.

V. CONCLUSION

We develop a user-friendly GUI for the molecular graphics software VMD, which allows researchers of theoretical simulations to conveniently translate and rotate the molecule

in the modeling process. This GUI is an open-source program, which can be obtained freely. Also, it can be freely modified to fit the specific needs of various VMD users.

AUTHOR INFORMATION

Corresponding Author

Zaixing Yang – Institute of Quantitative Biology and Medicine, State Key Laboratory of Radiation Medicine and Protection, School of Radiation Medicine and Protection, Collaborative Innovation Center of Radiological Medicine of Jiangsu Higher Education Institutions, Soochow University, Suzhou, Jiangsu 215123, China; orcid.org/0000-0003-3521-6867; Email: zxyang@suda.edu.cn

Authors

ChenChen Wu – Institute of Quantitative Biology and Medicine, State Key Laboratory of Radiation Medicine and Protection, School of Radiation Medicine and Protection, Collaborative Innovation Center of Radiological Medicine of Jiangsu Higher Education Institutions, Soochow University, Suzhou, Jiangsu 215123, China

Shengtang Liu – Institute of Quantitative Biology and Medicine, State Key Laboratory of Radiation Medicine and Protection, School of Radiation Medicine and Protection, Collaborative Innovation Center of Radiological Medicine of Jiangsu Higher Education Institutions, Soochow University, Suzhou, Jiangsu 215123, China

Shitong Zhang – Institute of Quantitative Biology and Medicine, State Key Laboratory of Radiation Medicine and Protection, School of Radiation Medicine and Protection, Collaborative Innovation Center of Radiological Medicine of Jiangsu Higher Education Institutions, Soochow University, Suzhou, Jiangsu 215123, China

Complete contact information is available at:
<https://pubs.acs.org/10.1021/acs.jcim.0c00754>

Notes

The authors declare no competing financial interest.

ACKNOWLEDGMENTS

We thank Dr. Ruhong Zhou, Xuanyu Meng, Xiuhua Yin, and Hong Zhou for helpful discussions. This work was partially supported by the National Natural Science Foundation of China (11574224 and U1967217), Collaborative Innovation Center of Radiological Medicine of Jiangsu Higher Education Institutions, Jiangsu Provincial Key Laboratory of Radiation Medicine and Protection, and A Project Funded by the Priority Academic Program Development of Jiangsu Higher Education Institutions (PAPD).

REFERENCES

- (1) Humphrey, W.; Dalke, A.; Schulten, K. VMD: Visual Molecular Dynamics. *J. Mol. Graphics* **1996**, *14*, 33–38.
- (2) Colletier, J. P.; Laganowsky, A.; Landau, M.; Zhao, M. L.; Soriaga, A. B.; Goldschmidt, L.; Flot, D.; Cascio, D.; Sawaya, M. R.; Eisenberg, D. Molecular Basis for Amyloid-Beta Polymorphism. *Proc. Natl. Acad. Sci. U. S. A.* **2011**, *108*, 16938–16943.
- (3) Ma, D. J.; Tillman, T. S.; Tang, P.; Meirovitch, E.; Eckenhoﬀ, R.; Carnini, A.; Xu, Y. NMR Studies of a Channel Protein without Membranes: Structure and Dynamics of Water-Solubilized KcsA. *Proc. Natl. Acad. Sci. U. S. A.* **2008**, *105*, 16537–16542.
- (4) Berneche, S.; Roux, B. Molecular Dynamics of the KcsA K⁺ Channel in a Bilayer Membrane. *Biophys. J.* **2000**, *78*, 2900–2917.

- (5) Egwolf, B.; Roux, B. Ion Selectivity of the KcsA Channel: A Perspective from Multi-Ion Free Energy Landscapes. *J. Mol. Biol.* **2010**, *401*, 831–842.
- (6) Lomize, A. L.; Pogozheva, I. D.; Lomize, M. A.; Mosberg, H. I. Positioning of Proteins in Membranes: A Computational Approach. *Protein Sci.* **2006**, *15*, 1318–1333.
- (7) Kimmitt, T.; Smith, N.; Witham, S.; Petukh, M.; Sarkar, S.; Alexov, E. ProBLM Web Server: Protein and Membrane Placement and Orientation Package. *Comput. Math. Methods Med.* **2014**, *2014*, 838259.
- (8) Lee, J.; Cheng, X.; Swails, J. M.; Yeom, M. S.; Eastman, P. K.; Lemkul, J. A.; Wei, S.; Buckner, J.; Jeong, J. C.; Qi, Y. F.; Jo, S.; Pande, V. S.; Case, D. A.; Brooks, C. L.; MacKerell, A. D.; Klauda, J. B.; Im, W. CHARMM-GUI Input Generator for NAMD, GROMACS, AMBER, OpenMM, and CHARMM/OpenMM Simulations Using the CHARMM36 Additive Force Field. *J. Chem. Theory Comput.* **2016**, *12*, 405–413.
- (9) Lomize, M. A.; Pogozheva, I. D.; Joo, H.; Mosberg, H. I.; Lomize, A. L. OPM Database and PPM Web Server: Resources for Positioning of Proteins in Membranes. *Nucleic Acids Res.* **2012**, *40*, D370–D376.
- (10) Aksimentiev, A.; Sotomayor, M.; Wells, D. Membrane Proteins Tutorial. Available at: <http://www.ks.uiuc.edu/Training/Tutorials/science/membrane/mem-tutorial.pdf> (accessed August 23, 2020).
- (11) Phillips, J. C.; Braun, R.; Wang, W.; Gumbart, J.; Tajkhorshid, E.; Villa, E.; Chipot, C.; Skeel, R. D.; Kale, L.; Schulten, K. Scalable Molecular Dynamics with NAMD. *J. Comput. Chem.* **2005**, *26*, 1781–1802.
- (12) Abraham, M. J.; Murtola, T.; Schulz, R.; Páll, S.; Smith, J. C.; Hess, B.; Lindahl, E. GROMACS: High Performance Molecular Simulations through Multi-Level Parallelism from Laptops to Supercomputers. *SoftwareX* **2015**, *1–2*, 19–25.
- (13) Huang, J.; MacKerell, A. D., Jr CHARMM36 All-Atom Additive Protein Force Field: Validation Based on Comparison to NMR Data. *J. Comput. Chem.* **2013**, *34*, 2135–2145.
- (14) Jorgensen, W. L.; Chandrasekhar, J.; Madura, J. D.; Impey, R. W.; Klein, M. L. Comparison of Simple Potential Functions for Simulating Liquid Water. *J. Chem. Phys.* **1983**, *79*, 926–935.
- (15) Stallman, R. <https://www.gnu.org/licenses/gpl-3.0.en.html> (accessed Aug. 23, 2020).